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1.

(a) Correct — AND gate followed by NOT gate with inputs 1, 1 produces 0.
(b) Faulty — 4-input AND gate with inputs 1, 1, 1, 0 should output 0, but shows 1.
(c) Faulty — OR gate with inverted inputs 0, 0 should output 1, but shows 0.
(d) Correct — 4-input NOR gate with inputs 0, 0, 0, 1 produces 0.
(e) Faulty — XOR gate with inputs 0, 1 should output 1, but shows 0.
(f) Correct — XOR gate with inputs 1, 1 produces 0.

2.

(a) Circuit = AND→NOT = NAND(A,B). Measured X tracks **B exactly, in every segment**, completely ignoring A (e.g., when A=1,B=1 → NAND should be 0, but X=1; when A=0,B=0 → NAND should be 1, but X=0). **Faulty** : output behaves like a buffer for B, not NAND(A,B).

(b) Circuit = OR→NOT = NOR(A,B). Measured X tracks A exactly (a repeating pulse train) for the *entire* diagram, even after B goes and stays HIGH — at which point NOR(A,B) must be constant 0 regardless of A, but X keeps pulsing just like A. **Faulty.**

(c) **Faulty** - OR gate with inverted inputs (Negative-OR / NAND equivalent) fails to output the correct waveform.
(d) **Faulty** - XOR gate fails to drop to LOW (0) when both inputs A and B are HIGH (1).

3.

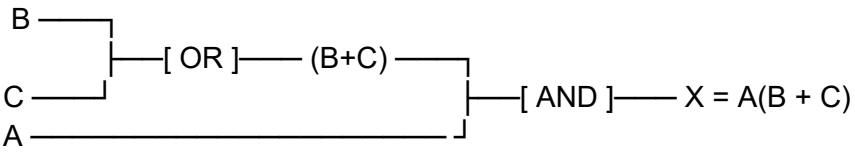
(a) (A + B)' = A'B'
(b) (A'B)' = A + B'
(c) (A + B + C)' = A'B'C'
(d) (ABC)' = A' + B' + C'
(e) [A(B + C)]' = A' + B'C'
(f) (AB + CD)' = (A' + B')(C' + D')
(g) (A'B + CD)' = (A + B')(C' + D')
(h) [(A + B')(C' + D)]' = A'B + CD'

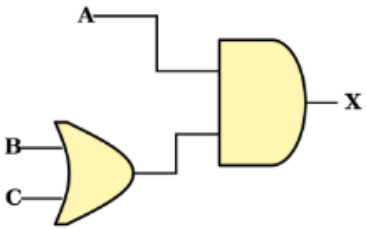
4.

(a) X = AB + C
(b) X = (A'B)'
(c) X = ABCD
(d) X = (A + B + C)D

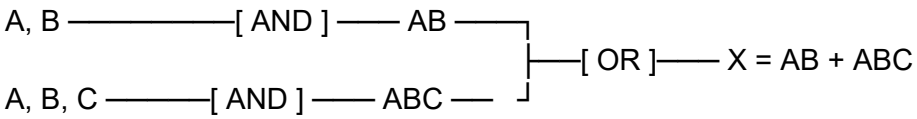
5

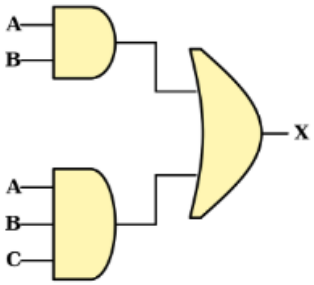
a) A(B+C)



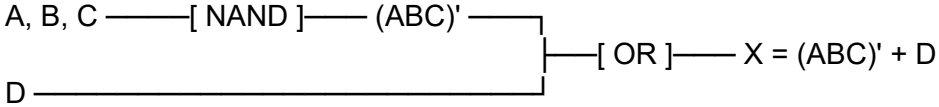


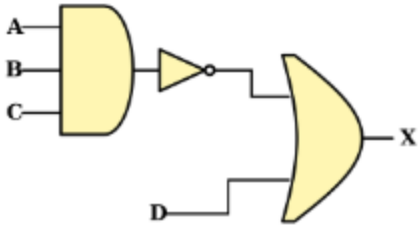
b) AB + ABC



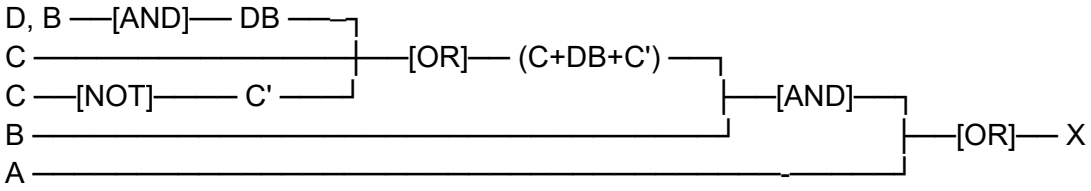


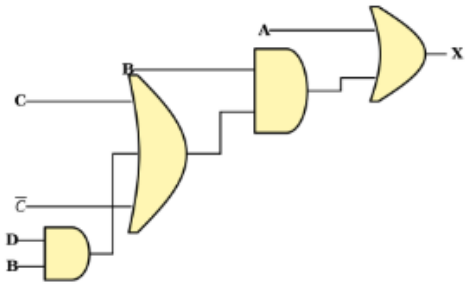
c) (ABC)’ + D





d) A + B(C + DB + C')





6.

(a) X = A + B

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

(b) X = AB

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

(c) X = A + B(C + D)

A	B	C	D	X
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

(d) X = A + B + C

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

(e) X = AB + BC

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0

1 | 0 | 1 | 0
1 | 1 | 0 | 1
1 | 1 | 1 | 1

7. Determine which of the logic circuits are equivalent.

(a) Step-by-Step Circuit Derivation:

- Gate 1 (3-input AND gate):

Takes inputs A, B, and C to output ABC.

- Gate 2 (2-input OR gate):

Takes output ABC from Gate 1 and input D to output (ABC + D).

- Gate 3 (2-input AND gate):

Takes output (ABC + D) from Gate 2 and input A from below to output A(ABC + D).

- Gate 4 (2-input AND gate):

Takes inputs B and C from below to output BC.

- Gate 5 / Output Gate (2-input OR gate):

Combines the outputs of Gate 3 and Gate 4 to produce
X = A(ABC + D) + BC

Boolean Simplification:
- Expand Gate 3: A(ABC + D) = A·ABC + AD = ABC + AD
- Combined **X = ABC + AD + BC**
- Using Boolean absorption rule (BC + ABC = BC):
  X = AD + BC

Final Output Expression for (a): **X = AD + BC**

(b) Step-by-Step Circuit Derivation:

- Gate 1 (2-input AND gate):

Takes inputs A and B to output AB.

- Gate 2 (2-input AND gate):

Takes inputs A and C to output AC.

- Gate 3 (2-input AND gate):

Takes inputs A and D to output AD.

- Gate 4 / Output Gate (3-input OR gate):

Combines the outputs of Gate 1, Gate 2, and Gate 3 to produce X:
X = AB + AC + AD

Boolean Simplification /:
- Factoring out A: X = A(B + C + D)

Final Output Expression for (b): **X = AB + AC + AD (or A(B + C + D))**

(c) Step-by-Step Circuit Derivation:

- Gate 1 (3-input AND gate):

Takes inputs A, B, and C to output ABC.

- Gate 2 (2-input AND gate):

Takes output ABC (from Gate 1) and input D to output ABCD.

- Gate 3 (2-input AND gate):

Takes input A AND input B to output AB.

- Gate 4 / Output Gate (2-input OR gate):

Combines output (ABC)D from Gate 2 and output AB from Gate 3:
X = ABCD + AB

Boolean Simplification:

Factor out AB: X = AB(CD + 1)
- Since (CD + 1) = 1: X = AB(1) = AB Final Output Expression for: **X = AB**

(d) Step-by-Step Circuit Derivation:

1. Gate 1 (2-input AND gate):
Takes inputs A and B to output AB.

2. Gate 2 (3-input AND gate):
Takes inputs A, C, and D to output ACD.

3. Gate 3 / Output Gate (2-input OR gate):
Combines output AB from Gate 1 and output ACD from Gate 2 to produce : X = AB + ACD

Boolean Simplification / Factoring:
- Factoring out A: X = A(B + CD)

Final Output Expression for (d): **X = AB + ACD (or A(B + CD))**

Conclusion:
None of the logic circuits are equal to each other because every circuit gives a different output expression.

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(a) X = ABC + A'BC + ABC'

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

(b) Output = XYZ + XY'Z + X'YZ + X'YZ' + XY'Z'

Truth Table:

X	Y	Z	Output
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

9. For each truth table in Table , derive a standard SOP and a standard POS expression.

(a) Standard SOP and POS Expressions:

Standard SOP (Sum-of-Products) [Where X = 1]:
- Row 001: A'B'C
- Row 100: AB'C'
- Row 101: AB'C
- Row 111: ABC

SOP Expression:
X = A'B'C + AB'C' + AB'C + ABC

Standard POS (Product-of-Sums) [Where X = 0]:

- Row 000: (A + B + C)
- Row 010: (A + B' + C)
- Row 011: (A + B' + C')
- Row 110: (A' + B' + C)

POS Expression: X = (A + B + C)(A + B' + C)(A + B' + C')(A' + B' + C)

(b) Standard SOP and POS Expressions:

Standard SOP (Sum-of-Products) [Where X = 1]:
- Row 101: AB'C
- Row 110: ABC'
- Row 111: ABC

SOP Expression: **X = AB'C + ABC' + ABC**

Standard POS (Product-of-Sums) [Where X = 0]:

- Row 000: (A + B + C)
- Row 001: (A + B + C')
- Row 010: (A + B' + C)
- Row 011: (A + B' + C')
- Row 100: (A' + B + C)

POS Expression: X = (A + B + C)(A + B + C')(A + B' + C)(A + B' + C')(A' + B + C)

(c) Standard SOP and POS Expressions:

Standard SOP (Sum-of-Products) [Where X = 1]:
- Row 0000: A'B'C'D'
- Row 0001: A'B'C'D
- Row 0011: A'B'CD
- Row 0101: A'BC'D
- Row 0110: A'BCD'
- Row 1001: AB'C'D
- Row 1100: ABC'D'

SOP Expression:
X = A'B'C'D' + A'B'C'D + A'B'CD + A'BC'D + A'BCD' + AB'C'D + ABC'D'

Standard POS (Product-of-Sums) [Where X = 0]:

- Row 0010: (A + B + C' + D)
- Row 0100: (A + B' + C + D)
- Row 0111: (A + B' + C' + D')
- Row 1000: (A' + B + C + D)
- Row 1010: (A' + B + C' + D)
- Row 1011: (A' + B + C' + D')
- Row 1101: (A' + B' + C + D')
- Row 1110: (A' + B' + C' + D)
- Row 1111: (A' + B' + C' + D')

POS Expression: X = (A + B + C' + D)(A + B' + C + D)(A + B' + C' + D')(A' + B + C + D)(A' + B + C' + D)(A' + B + C' + D')(A' + B' + C + D')(A' + B' + C' + D)(A' + B' + C' + D')

(d) Standard SOP and POS Expressions:

Standard SOP (Sum-of-Products) [Where X = 1]:
- Row 0010: A'B'CD'
- Row 0100: A'BC'D'
- Row 0101: A'BC'D
- Row 0111: A'BCD
- Row 1011: AB'CD
- Row 1100: ABC'D'
- Row 1111: ABCD

SOP Expression:
X = A'B'CD' + A'BC'D' + A'BC'D + A'BCD + AB'CD + ABC'D' + ABCD

Standard POS (Product-of-Sums) [Where X = 0]:

- Row 0000: (A + B + C + D)
- Row 0001: (A + B + C + D')
- Row 0011: (A + B + C' + D')
- Row 0110: (A + B' + C' + D)
- Row 1000: (A' + B + C + D)
- Row 1001: (A' + B + C + D')
- Row 1010: (A' + B + C' + D)
- Row 1101: (A' + B' + C + D')
- Row 1110: (A' + B' + C' + D)

POS Expression: X = (A + B + C + D)(A + B + C + D')(A + B + C' + D')(A + B' + C' + D)(A' + B + C + D)(A' + B + C + D')(A' + B + C' + D)(A' + B' + C + D')(A' + B' + C' + D)

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(a) Expression: A'B'C + AB'C + A'BC'

*. K-Map Plotting (Rows = A, Columns = BC):

```
          BC
        00  01  11  10
       +----------------
A   0 |  0 | 1 |   0  | 1 |   (A=0: A'B'C at 001, A'BC' at 010)
    1 |  0 | 1 |   0  | 0 |   (A=1: AB'C at 101)
```

*. Grouping:
- Group 1 (Vertical pair at BC = 01): Combines A'B'C and AB'C
  --> Eliminates A, leaves B'C
- Group 2 (Single 1 at A=0, BC=10): A'BC' (cannot be grouped)
  --> Remains A'BC'

Minimum SOP Expression for (a): X = B'C + A'BC'

(b) Expression: AC(B' + C)

1. Expansion into SOP form:
- AC(B' + C) = AB'C + ACC
- Since CC = C: AB'C + AC
- Since AC + AB'C = AC(1 + B') = AC:
  The simplified expression directly equals AC.

* K-Map Plotting (Rows = A, Columns = BC):
  - Minterm AC corresponds to cells where A = 1 and C = 1 (101 and 111):
    AB'C = 101
    ABC = 111

```
          BC
       00  01  11  10
     +----------------
A  0 |  0 | 0 | 0 | 0 |
   1 |  0 | 1 | 1 | 0 |   (A=1: AB'C at 101, ABC at 111)
```

* Grouping:
  - Group 1 (Horizontal pair at A = 1, BC = 01 & 11):
    Combines AB'C and ABC.
    --> Eliminates B, leaves AC.

Minimum SOP Expression for (b): X = AC

(c) Expression: A'(BC + BC') + A(BC + BC')

* Expansion / Simplification:
  - Notice that both terms share the factor (BC + BC'):
    (A' + A)(BC + BC')
  - Since (A' + A) = 1:
    X = BC + BC'
  - Factoring out B:
    X = B(C + C') = B(1) = B

* K-Map Plotting (Rows = A, Columns = BC):
  - SOP terms: A'BC, A'BC', ABC, ABC' (Row 0: 011, 010 | Row 1: 111, 110)

```
          BC
       00  01  11  10
     +----------------
A  0 |  0 | 0 | 1 | 1 |   (A=0: A'BC at 011, A'BC' at 010)
   1 |  0 | 0 | 1 | 1 |   (A=1: ABC at 111, ABC' at 110)
```

* Grouping:
  - Group 1 (Quad of four 1s forming a 2x2 block across columns BC = 11 and 10):
    Combines A'BC, A'BC', ABC, and ABC'.
    --> Eliminates both A and C, leaving B.

Minimum SOP Expression for (c): X = B

(d) Expression: A'B'C + AB'C' + A'BC' + ABC'

* Minterm Identification:
  - A'B'C = 001
  - AB'C' = 100
  - A'BC' = 010
  - ABC'   = 110

* K-Map Plotting (Rows = A, Columns = BC):

```
          BC
       00  01  11  10
     +----------------
  A 0 |  0 | 1 | 0 | 1 |   (A=0: A'B'C at 001, A'BC' at 010)
    1 |  1 | 0 | 0 | 1 |   (A=1: AB'C' at 100, ABC' at 110)
```

* Grouping:
  - Group 1 (Horizontal pair in Row 1 at BC = 00 & 10):
    Combines AB'C' (100) and ABC' (110).
    --> Eliminates B, leaves AC'

  - Group 2 (Vertical pair at BC = 10 across Row 0 & Row 1):
    Combines A'BC' (010) and ABC' (110).
    --> Eliminates A, leaves BC'

  - Group 3 (Single 1 at A=0, BC=01):
    A'B'C (001) cannot be grouped with any adjacent 1s.
    --> Remains A'B'C

Minimum SOP Expression for (d): X = AC' + BC' + A'B'C